

Class 10 Science — Chapter 1: Chemical Reactions and Equations

Detailed Revision Notes

1. Chemical Reactions & Observations

A chemical reaction involves a chemical change where original substances lose their nature and identity to form new substances with completely different properties.

Four key observations that help determine if a chemical reaction has taken place:

- Change in state
- Change in colour
- Evolution of a gas
- Change in temperature

2. Chemical Equations & Balancing

A chemical equation is a symbolic representation of a chemical reaction using symbols and formulae of the substances involved.

Law of Conservation of Mass: Mass can neither be created nor destroyed in a chemical reaction. Therefore, the total number of atoms of each element must remain equal on both the reactant and product sides (Balanced Equation).

3. Types of Chemical Reactions

Reaction Type	Description	General Formula / Example
Combination	Two or more reactants combine to form a single product.	$A + B \rightarrow AB$ $C + O_2 \rightarrow CO_2$
Decomposition	A single reactant breaks down into two or more simpler products. Requires energy (Thermal, Electrolytic, or Photolytic).	$AB \rightarrow A + B$ $2H_2O \rightarrow 2H_2 + O_2$
Displacement	A more reactive element displaces a less reactive element from its salt solution.	$A + BC \rightarrow AC + B$ $Fe + CuSO_4 \rightarrow FeSO_4 + Cu$

Double Displacement	Two compounds exchange ions to form two new compounds, often forming a precipitate.	$AB + CD \rightarrow AD + CB$
Oxidation & Reduction (Redox)	Oxidation is the gain of oxygen or loss of hydrogen. Reduction is the loss of oxygen or gain of hydrogen.	$CuO + H_2 \rightarrow Cu + H_2O$

4. Effects of Oxidation in Daily Life

Corrosion: The gradual eating up of metals by the action of air, moisture, or chemicals on their surface (e.g., Rusting of iron: reddish-brown coat).

Rancidity: The oxidation of fats and oils in food item material kept for a long time, leading to unpleasant smells and tastes. Prevention includes adding antioxidants or flushing packages with Nitrogen gas.